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ORIGINAL RESEARCH

# A Multi-Level Blockchain-Based Architecture for Securing and Controlling Data Flow in 6G Networks

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## Abstract

In 6G networks, the computing and storage capabilities of smart devices will experience significant growth. By 2030, it is expected that hundreds of billions of devices will be interconnected worldwide. Future systems will need to support the explosive growth of data traffic, providing ultra-high speeds and ultra-low latency. These stringent requirements demand robust architectures capable of ensuring both performance and security. The increase in data flow within 6G networks necessitates robust security solutions. This paper introduces a cuttingedge architecture that leverages the strengths of blockchain and fog computing to enhance security. The proposed architecture is further integrated with Mobile Edge Computing (MEC) to form a resilient Vehicular Edge Computing (VEC) framework. This VEC system addresses several critical requirements of 6G networks, including the secure and efficient management of resources and data flows. The paper explores a lightweight multi-user authentication scheme for secure vehicular networks, leveraging fog computing and blockchain technology. The proposed solution addresses critical security challenges while maintaining efficient communication within the vehicular ecosystem.

**Keywords** 6G · Blockchain · Mobile edge computing (MEC) · Vehicular edge computing (VEC) · Smart contracts · Connected vehicles · Low latency · Resource management

## Introduction

Traditional networks are unable to meet the requirements of modern computing. The transformation of mobile communication technology began with 1G cellular networks in the 1980s. Since then, significant developments have been integrated into the telecommunications industry to address the challenges of each network generation (1G, 2G, 3G, 4G, and 5G) [1]. While the world has yet to benefit fully from 5G coverage [2], researchers are already investigating the network future, often called 6G. Nhu [3] expect that the roll-out plan of commercializing 6G will be in 2030. Research collaborations are initiating investigations of guidelines that will shape the vision of 6G [4]. Smart applications in 6G environments are exposed to several security vulnerabilities due to their decentralized nature and the massive increase in the number of connected devices. These vulnerabilities include unauthorized access through security holes in IoT devices to access critical systems. As well as increased data loss due to the transfer of large amounts of sensitive information between nodes. Man-in-the-Middle (MITM) attacks also pose a threat to these applications as they can intercept and manipulate vehicle-to-vehicle (V2X) communications. Additionally, 6G networks are vulnerable to denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks, which can disrupt services by overloading resources. However, data corruption is a major risk because without robust traceability, the integrity of the information exchanged can

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be compromised, affecting the reliability of these applications. Previous generations of mobile networks faced several challenges, including bandwidth issues, latency concerns, privacy and security issues, and authorization and authentication issues. Hence, the security vulnerabilities have evolved. Moreover, the increasingly massive data flow is a substantial challenge. It is driven by the growth in wireless communication of smart Internet of Things (IoT) devices and powerful mobile applications that significantly enhance daily life [5, 6]. Implementing these applications requires high-speed and low-latency data transmission. However, links between different nodes are consistently limited by bandwidth, making it difficult to meet transmission requirements. This data-intensive characteristic poses significant challenges for mobile networks, especially in scenarios involving many smart devices. ME) caching addresses these challenges by alleviating the link load. It benefits from caching resources at Edge nodes and provides access to popular content near end-users. The necessity to deploy MEC arises as a promising paradigm that brings computing resources to mobile users at the network edge [7]. MEC enables computationally demanding and latency-sensitive applications to be swiftly processed by edge servers, meeting the requirements of mobile users. Considering new telecommunication systems across various domains such as health, transportation, climate, and cybersecurity is required for security. Furthermore, securing data become a critical issue in mobile applications. As terminals evolve, the collected data may contain sensitive information from many users. Integrating MEC into 6G requires addressing user privacy and security issues. Due to MEC systems' high dynamism and openness, reliably and securely safeguarding information regarding exchanges, traceability, and service/billing from edge servers and all possible data flows becomes highly challenging. Blockchain technology presents an optimal solution to enhance the security and privacy of 6G systems by leveraging its numerous advantages and promising techniques, including decentralization, privacy, immutability, traceability, and transparency. The convergence of Blockchain and MEC in 6G systems could enable secure orchestration and performance enhancements. Leveraging the smart contract paradigm facilitates the automation of conventions and processes, ensuring the security of all exchanges and communications within such a system.

Despite these advances in mobile networks, several challenges remain, including security, efficient resource management, and latency reduction for real-time communications. Existing solutions fail to provide robust security against new threats, such as attacks on V2X communications or secure transaction management in a highly dynamic environment. Moreover, traditional cloud-based architectures often incur high latency due to the distance between devices and processing centers. This paper proposes an innovative architecture, based on the integration of blockchain and VEC, that aims to address these limitations by ensuring secure and efficient data management with minimal latency. This approach helps address critical needs of 6G networks, such as secure resource management and transaction automation via smart contracts. The study aims to address the two following aspects:

- Intensive computation with ultra-low latency in MEC environment.
- Security strategies must be robust to manage large data flows and Vehicular to-everything (V2X) communications with inadequate centralized authentication systems.

The proposed approach aims to enhance the network performance and reduces the network latency in V2X communications through MEC environment. We also use Blockchain technology for secure and efficient data handling and authentication in 6G networks. Consequently, the Blockchain architecture is integrated with Service Level Agreement (SLA)-based smart contracts for secure transactions and real-time billing.

The remainder of this paper is organized as follows: "Related Work" section reviews related works, highlighting the limitations of current technologies and the potential of Blockchain and MEC for enhancing security and performance. "VEC: A Blockchain and MEC-Based System for Reliable and Secure V2X Communications" section introduces the VEC system, a novel architecture using Blockchain and smart contracts for secure and efficient V2X communications, detailing its design and security measures. "Use Cases" section depicts the use cases of our architecture. "Implementation Scenarios" section describes implementation scenarios. "Discussion and Challenges" section provides a discussion and challenges of our proposal. Finally, the "Conclusion" section concludes our work.

## Related Work

The V2X communication plays a fundamental role in developing Intelligent Transportation Systems (ITS). The integration of cloud technologies into vehicular networks opens new ways to optimize traffic management [8], strengthen road safety [9], propose proactive logistics services [10] and improve traffic simulation [11, 12]. V2X architectures are based on V2V, V2I communications, and more broadly, towards all elements that can influence driving (V2X), including pedestrians, networks, and cloud services (V2C) [13]. Network protocols such as virtual routing [2] and C-V2X, enriched by the move to 5G, provide reduced latency and increased transmission capacity, which is crucial for efficiently deploying ITS services. These architectures

encounter constraints that may impact their overall effectiveness and performance, including high latency, inconsistent connectivity, sensitivity to bandwidth, security, and data privacy concerns. Sakraoui et al. [14], propose a secure gradient swapping algorithm for intrusion detection in 6G networks, leveraging federated learning, secure multiparty computation, and blockchain. The approach aims to ensure data confidentiality while optimizing communication and computation costs. Edge Computing was introduced to address the need for low-latency processing, a requirement that traditional edge infrastructures struggle to meet. Additionally, edge computing offers scalability to accommodate the surge in connected vehicles, which conventional Cloud infrastructures find challenging. Several software solutions deployed for Edge Computing operate in a Cloud infrastructure and vice versa. Nowadays, Edge infrastructures are proposed for future 6G networks to provide low latency and substantial computing power for reliable communication in IoT systems [15]. However, it is noteworthy that numerous MEC devices openly share their resources or content, often overlooking privacy concerns. Integrating Blockchain could secure MEC systems [16]. The escalating growth of the Internet of Things (IoT) and wireless technologies has led to an unprecedented surge in data and content volumes. Addressing this challenge, MEC emerges is a pivotal solution that provides distributed computing and caching resources near users. MECs allow efficient processing and caching of content at the network edge, mitigating data traffic on terrestrial links and reducing content delivery latency. Furthermore, MEC techniques find applications in diverse scenarios, catering to intensive computation, processing, and analysis demands in domains like disease prevention, industrial production, and emergency interventions [17]. IoT applications often function as sophisticated monitoring systems, collecting and analyzing data, and triggering actions based on generated insights [18]. Here, Edge Computing proves advantageous for real-time information needs by bringing computation closer to IoT devices, reducing network latency through shorter round trips to data centers. This optimization is crucial for IoT applications requiring prompt actions. Additionally, MEC addresses bandwidth considerations, as the cumulative effect of small data packets broadcast in real time from numerous devices can be significant. Edge Computing facilitates data processing and filtering closer to devices, optimizing bandwidth by transmitting only essential data for storage or longer-term analysis to centralized management platforms [19]. Edge Computing offers security advantages for enterprises concerned about data storage in locations subject to different protection laws. When Edge servers are on-site, companies can ensure data remains within their local perimeter, exercising precise control over access to servers storing sensitive information [20]. Maksymyuk et al. [21] integrate blockchain mobile network infrastructure. The authors consider a tokenization model for spectrum and infrastructure, examining its distributed ledger structure and suitable consensus algorithms. Therefore, they implement and analyze smart contracts for semi-persistent, dynamic, and intelligent spectrum trading, assessing transaction volumes for each method. The works of [22] and [4] use blockchain technologies to enable resource sharing and management in 6G networks. The authors discussed several application domains for integrating Blockchain for monitoring and managing data, including the Internet of Things, device-to-device communications, network slicing, and inter-domain blockchain ecosystems. Shen et al. [23] investigate blockchain technologies in 6G networks within three aspects of decentralized data management—first, consensus protocols, scalability mechanisms, and the role of data management stakeholders in blockchain architectures. Second, the techniques of authentication and authorization, and finally, preserving privacy in 6G networks using Blockchain. Hu et al. [24] use blockchains and Artificial Intelligence (AI) to dynamically share resources in wireless networks. AI is used to improve decision-making, while blockchains are used to preserve privacy. Several surveys and reviews like in [25–31] investigate the use of blockchains in 6G networks.

## VEC: A Blockchain and MEC-Based System for Reliable and Secure V2X Communications

Our proposal uses blockchain technology to secure V2X communications. We integrate Blockchain with MEC, leveraging Smart Contracts to ensure and enhance data security and guarantee non-repudiation of transactions concerning Vehicle/Edge infotainment exchanges. The system aims to provide secure and reliable functionalities for billing, payment, refunds, evaluations, and ensuring data non-falsifiability.

The proposed system architecture integrates Blockchain and MEC to enhance data security and control in 6G networks, specifically within VEC. It focuses on providing computational and storage solutions, infotainment services, and secure toll transactions (see Fig. 1). Blockchain is employed for secure and efficient data processing and authentication, while MEC reduces latency in V2X communications. The architecture also includes Smart Contracts for secure transactions and real-time billing, showcasing a commitment to security, flexibility, and performance enhancement in 6G networks.

Indeed, with the rapid evolution of technology, the demand for cutting-edge security solutions and stringent data control protocols in 6G networks has become increasingly pressing. The proposed VEC system will leverage Blockchain and MEC technologies to address these




**Fig. 1** Four-level architecture for V2X communication

challenges effectively. Blockchain will secure transactions and data exchanges, while MEC will expedite processing at the network edge, thereby reducing latency and enhancing reliability.

## VEC Proposal Architecture

The proposed VEC system is based on a multi-level architecture to ensure better consideration of security and Qos. Our system is hosted on four layers, as shown in Fig. 2. Each layer contains several components as in the following:

1. **Cloud layer:** The cloud contains blockchains containing the necessary information about clients and providers. Information such as payment methods, services, and history of QoS are included.
2. **Fog layer:** The fog contains three components: a service mediator, blockchains for saving smart contacts, and a service locator.
3. **Service mediator:** This service is responsible for linking clients to providers. It controls whether the service is executed as expected and suspends or terminates it when necessary. The mediator is also responsible for applying a handover mechanism with the help of the edge layer.
4. **Blockchains of smart contracts:** Each block contains the provider, the client, the price, and the terms of the contracts.
5. **Service locator:** discovery and select services for clients.
6. **Edge layer:** The Edge is responsible for communication between clients from one side and fog and cloud layers from another. It also contributes to applying the handover mechanism. Therefore, the client will benefit from the service when moving.
7. **End layer:** This layer contains a service monitor that collaborates with the mediator and service provider during execution. It also has applications that request services by invoking a service locator from the fog layer.

The smart contracts enable authentic and automatic access for various entities, leveraging the capabilities of the forthcoming 6G generation, which supports high mobility for critical systems that require significant computational and storage capacities. This approach makes it possible to design reliable, highly mobile, secure, non-repudiable systems that are resource-intensive in terms of computational and storage resources. This is what we have aimed to achieve in our work.

Smart contracts are pivotal in the VEC system as they automate and enforce agreements between parties involved in vehicle/Edge infotainment exchanges. Smart contracts facilitate secure billing, payments, refunds, and evaluation processes. By leveraging smart contracts, stakeholders can streamline processes and enhance security and trust in all transactions.

The VEC system is designed to efficiently scale to meet the increasing demands and complexities of 6G networks over time. Our architecture ensures flexibility and adaptability by leveraging Blockchain and MEC. Blockchain's decentralized nature allows the seamless addition of new nodes, while MEC's edge computing capabilities distribute resources effectively. Continuous optimization strategies ensure the system can dynamically scale to handle the evolving needs of 6G networks, maintaining optimal performance without compromise.

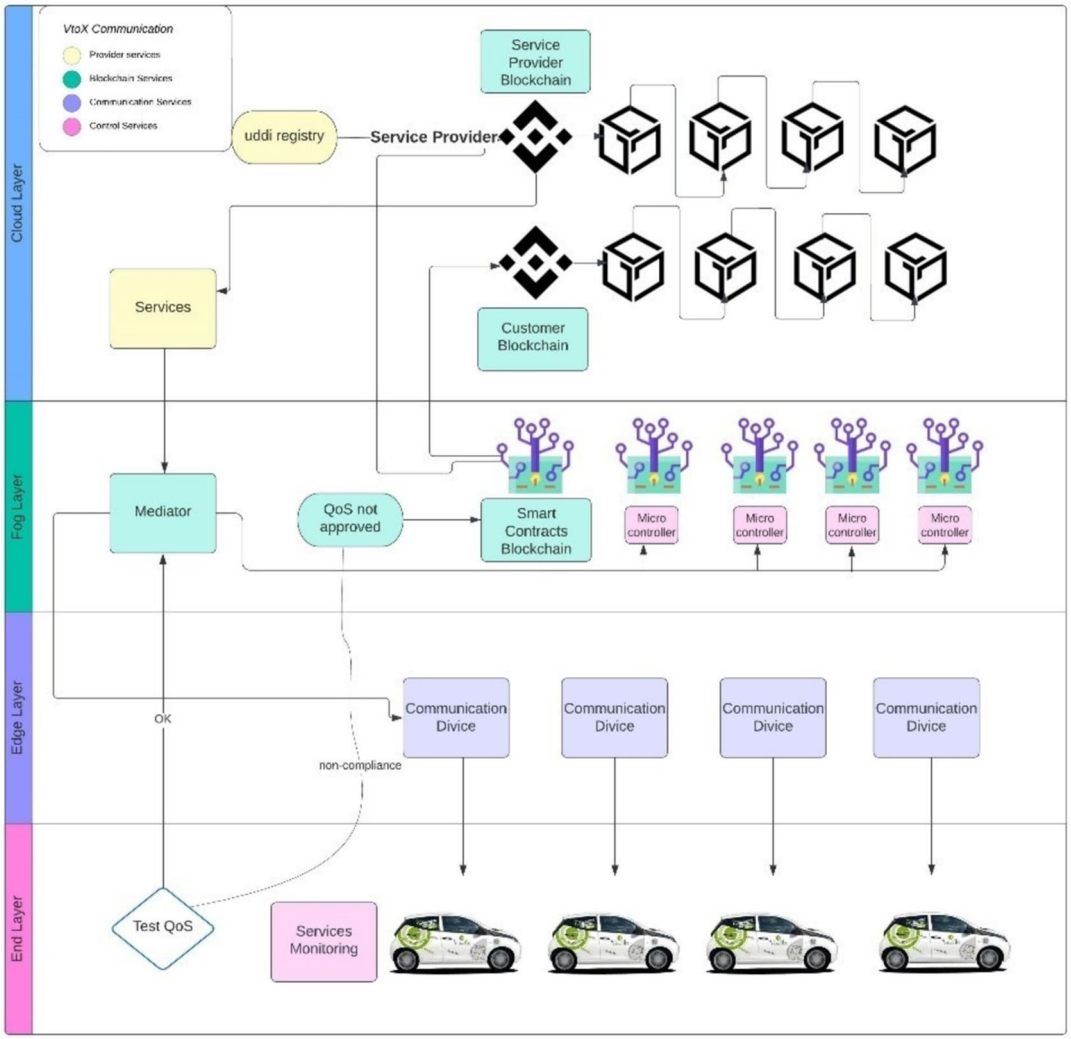



**Fig. 2** The architecture of V2X Communication System

## Security and V2X Technology

As cars become increasingly connected and autonomous driving becomes a reality, automotive manufacturers face a significant challenge: how to trust the information received by each vehicle. In this context, V2X communication will play a crucial role in the success of autonomous driving. To enhance data security and ensure the nonrepudiation of transactions related to the services provided by a vehicle, we propose the VEC system equipped with smart contracts. This system is based on the Ethereum Blockchain platform and utilizes its MetaMask test network. Given that the latter is a paid service, we leveraged its version Rinkeby, where all vehicles connected to this system can communicate with each other. They can also invoke Edge resources through a set of blockchains and smart contracts that guarantee data integrity and offer high security, billing, payment, and refunding, with non-falsification of transactions.

In most countries, governments consider V2X technology as a crucial means to explore fully the resources available at the Edge and reduce road accidents. They urge the industry to accelerate the deployment schedule of this technology. For automotive manufacturers, V2X technology, especially V2V, represents a milestone towards fully autonomous vehicles.

The MEC system is a technology-agnostic platform that enables deploying applications and resources closer to vehicles and pedestrians, thereby reducing packet transmission delay. The proposed MEC system will enhance existing V2X communications and play a significant role in V2X data exchanges. MEC enhances the ultra-low latency and high bandwidth characteristics of the proposed 6G solution by enabling edge processing, real-time decision-making, content caching, traffic offloading, and network slicing. By leveraging these capabilities, MEC accelerates data exchange, reduces latency, and optimizes resource

utilization, unlocking the full potential of V2X communications in the 6G era.

Our proposed solution brings computing closer to traffic and services from a centralized end-user Cloud. It facilitates data storage, processing, and data analysis at the network edge rather than in a remote data center, significantly reducing latency. Our Rinkeby blockchain sub-network ensures high security for V2X exchanges and communications. The integration of Blockchain technology, particularly on the Ethereum platform with Smart Contracts, establishes a robust framework for data security and transaction integrity in V2X communications. This integration ensures heightened data security through tamper-proof records and cryptographic protocols, guaranteeing the non-repudiation of transactions and ultimately fostering trust and reliability in V2X communications.

The suggested solution is characterized by ultra-low latency and high bandwidth, offering real-time access to radio network information that our application can exploit. Our MEC solution will play a significant role in future networks as it has the potential to support many latency-sensitive and real-time applications. The multi-level architecture of the proposed VEC system encompasses different layers that work synergistically to enable secure and automated access for various entities in the 6G networks. Thus, by incorporating Smart Contracts at different levels of the architecture, the VEC system enables secure and automatic access for entities in the 6G generation and enhances operational efficiency, transparency, and trust within the Vehicle/Edge infotainment ecosystem.

For instance, in the edge devices and vehicles layer, Smart Contracts enable secure data exchange between these devices and users. Smart Contracts oversee access control and service-level agreements among edge resources and service providers within the Edge Computing and Infotainment Services layer. Moving to Blockchain Integration,

Smart Contracts deployed on the Blockchain automate enforcement of agreements, ensuring secure billing and payment processing across all system entities. Additionally, in Secure and Automatic Access Control, Smart Contracts empower the system to verify that only authorized entities can access designated resources and services, thereby bolstering overall security and privacy measures.

## Scenario and Smart Contract Utilization Steps

Smart contracts are protocols that the system automatically uses to facilitate negotiations between contracting parties: the vehicle and the provider. Blockchain technology enhances the authentication of the latter, while contracts facilitate and secure blockchain transactions and the automatic execution of tasks triggered by contracts; thus, the two concepts are closely linked, as shown in the Fig. 3.

In this work, we have defined three Blockchains:

- **Provider blockchain** is used to register suppliers with their services and the details of each service from the point of view of quality, price, trust factor, etc.
- **Vehicle blockchain** records customers' details, namely the devices used, their account and profile, etc.
- **Transaction blockchain** contains the stakeholders of a transaction with the contract terms specifying the conditions of use of the services, the price, the duration, and the QoS.

The Fig. 4 shows the verification and execution steps of the smart contract.

We integrate the smart contract protocol in the following scenarios:

- We use it to specify the conditions (contract terms) for granting a paid service request from the requester (vehicle) to a provider (infotainment, toll pass, storage resources, computing resources, and others).
- Contract establishment occurs through term negotiation between the two parties; once the conditions are satisfied, the contract is established and recorded in a block, and the requested service is provided to the requester (vehicle).
- During service operation, if at least one condition is disrupted, an automatic contract cancellation is triggered, and a refund for the operation is executed to its requester.
- The service provider, having a parameter that specifies their trust level in service quality (Trust Level), will either maintain or improve this parameter by the client if the operation of a provided service is completed, or it will be lowered if the contract is canceled due to the unreliability or unavailability of the service.

### Dynamic Renegotiation of Smart Contracts

Dynamic renegotiation of smart contracts allows for automatic revision of initial conditions based on changing needs or available resources, without requiring human intervention. This approach ensures increased flexibility and optimal service continuity, especially in dynamic environments such as VEC networks. In the following, we will demonstrate the dynamic renegotiation process of smart contracts that allows the system to adapt quickly to unforeseen changes. This process thus ensures a high and continuous Qos. This flexibility is crucial in 6G networks, where the variability of conditions and the responsiveness of the system are essential.

The network conditions can change unpredictably, affecting current contracts. For example, a contract providing for a throughput of 100 Mbps between a vehicle and a provider may no longer be respected in the event of

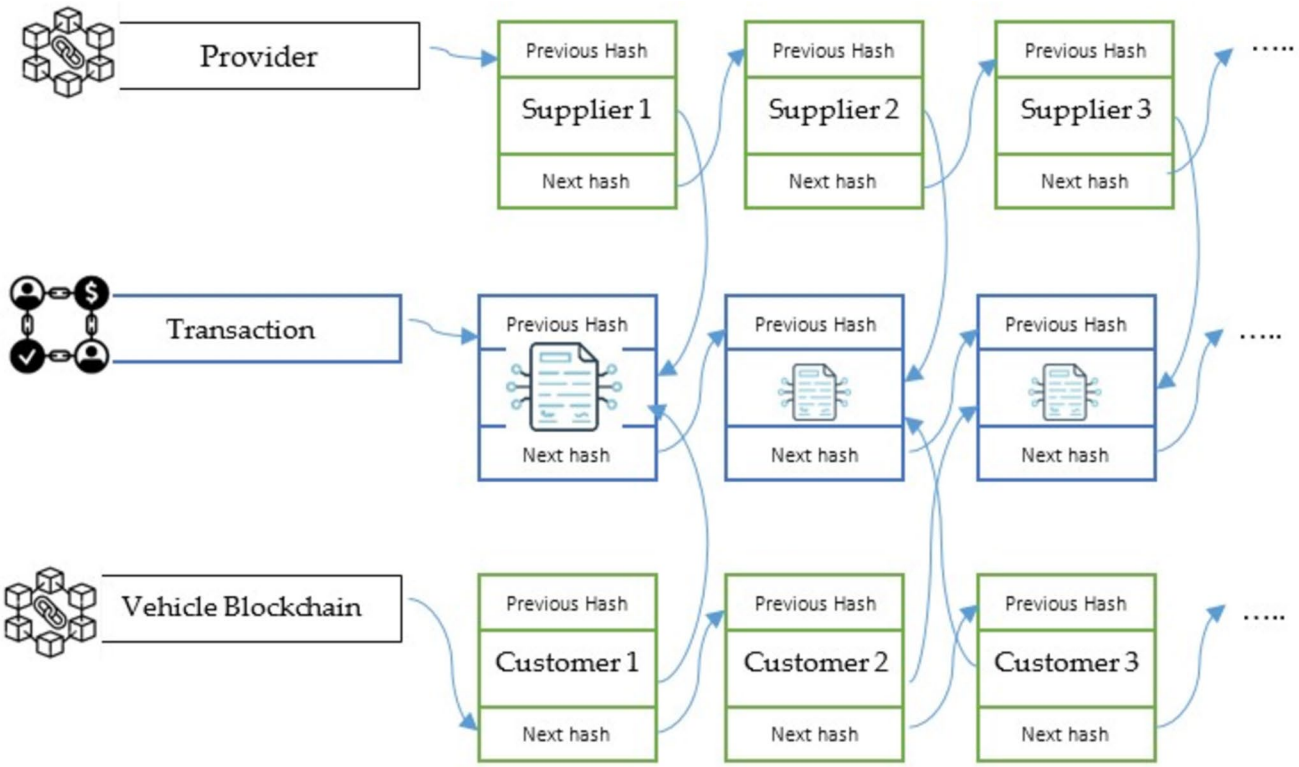



**Fig. 3** The blockchains used by our system

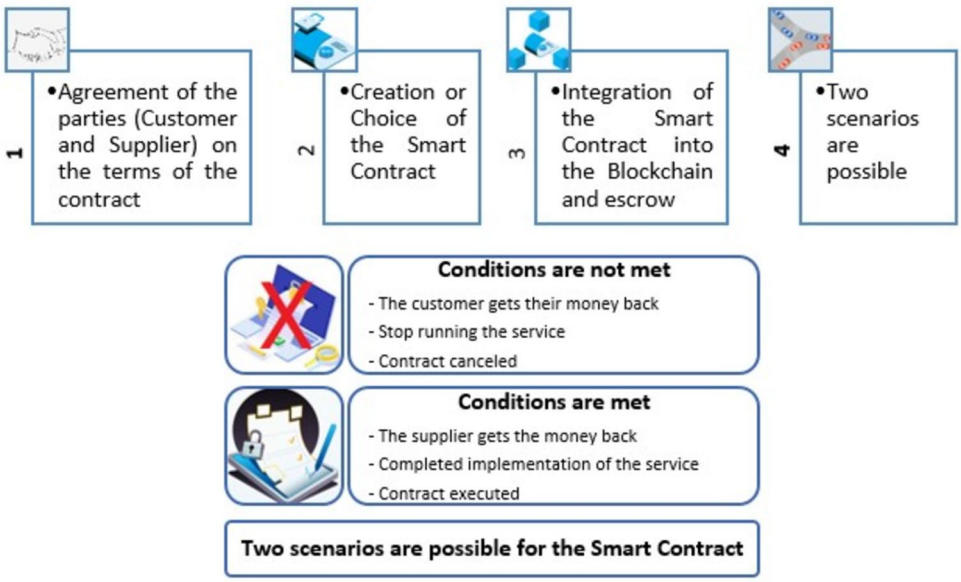



**Fig. 4** Life Cycle of a Smart Contract

network congestion. In this case, a dynamic renegotiation is triggered: the contract is adjusted by offering a reduced throughput with adapted pricing or a switch to a more efficient provider.

Another scenario concerns adaptation to user needs. If a user has initially reserved storage capacity on a MEC node for two hours, but wishes to extend the duration, the system can detect this extension and automatically renegotiate the contract. The new version adjusts the duration and cost according to the availability of resources, thus ensuring uninterrupted service continuity.

Finally, the management of unplanned interruptions is another use case for dynamic renegotiation. For example, in the event of a MEC node failure, the system immediately renegotiates a new contract with another available node. The original contract is cancelled, and an adjustment to the amount already paid is made to reflect the new benefit, ensuring a smooth and transparent transition.

### Driver Recommendation System

The recommendation system in our architecture is designed to improve the driving experience by proposing personalized services based on previous interactions, driver preferences, and real-time network conditions. The goal is to ensure a smooth and efficient driver experience, while taking into account network constraints and resource availability.

The data collection and analysis process is the first step. The system collects information about driver interactions with services, such as the type of service consumed, the duration of use, and the ratings provided. This data is securely stored in the blockchain. The architecture uses machine learning algorithms and deep learning models to identify trends in user preferences and anticipate their needs. If a driver frequently accesses storage services during peak hours in the road, the system can offer to reserve storage space in advance or recommend specific promotions.

The system also adapts its recommendations based on real-time network conditions. If a user wants to access a service through a currently overloaded VEC node, the recommendation system will automatically suggest a less congested alternative, ensuring service continuity without performance degradation. In addition, the system's flexibility is ensured by continuous feedback: each interaction, whether it is an acceptance or rejection of a recommendation, is analyzed to refine future suggestions. This allows the system to dynamically adjust to changing user needs and improve the accuracy of its recommendations over time.

## Processes and Algorithms of the VEC Platform

The platform encompasses a range of services describing the provided use cases and is distributed across four lanes of activity, concerning primarily the vehicle and its connection with the network, then the service provider; the latter is not provided except after negotiation and the establishment of a smart contract that will be executed at the edge level.

1. Once the connection is established, the vehicle is presented with a variety of services (free and paid) offered by providers with an initial trust level. The vehicle only needs to choose the provider from the list of choices and then invoke one of the services offered.
2. The selection of a service is necessarily followed by the negotiation of its characteristics, which must be agreed upon by both the vehicle and the provider to build a smart contract ensuring the proper use of resources. The service requester must specify several product criteria, including its type, quality, availability duration, price, etc.
3. Once the contract terms are specified and validated by both parties (requester, provider), the vehicle can transmit a contract establishment request to the Ethereum MetaMask network, which assigns it a block and a hash.
4. A paid service is always provided following a payment made during the contract establishment. The requester will be notified by a message indicating the amount debited from their account.
5. Once the contract establishment is completed, service exploration begins. Since it involves a smart contract, if at least one contract term is not fulfilled, a forced cancellation is automatically triggered, the vehicle is notified, and it can switch to another provider.
6. Case of terminating the Vehicle/Service Provider relationship:

    (a) In the event of a forced contract cancellation, the requester must be reimbursed, and an alert indicating the amount reimbursed to them is displayed.
    (b) The requester is notified after the end of service exploration (expiry of availability time).
    (c) The requester evaluates a provider based on how the contract is terminated; in the case of a forced cancellation, the Trust Level will be lowered, whereas it is maintained or improved in the expected termination of the contract.
    (d) After exploring the platform, certain disconnection is inevitable, which occurs following the vehicle's request.

The algorithm that requests and exploits resources comprises four phases:

- Vehicle connection
- Service search and choice
- Negotiation of contract terms
- Establishment of contract

**Algorithm 1** Service search and selection

```
Require: V: vehicle, S: service, BCV: Blockchain for connected vehicles
Listening platform ()
if receive connection dde (V) then
    Connect (V)
    Insert Block (BCV)
    Treat DDS (V, S)
end if
```

**Algorithm 2** Algorithm DDS(V,S)

```
F: Supplier
(2) View register (S)
Choose (S,F)
if (S is Found) then
    Negotiate (V, F, S)
    if Negotiate == True then
        Establish SmartContract (S, QoS, Time, Price)
        Use (V, S)
    else
        Go (2)
    end if
else
    Service not found ()
```

**Creation of a Block:** After selecting the provider and establishing a contract between the vehicle and the provider, a block will be added to the chain to preserve the rights of clients (i.e., the vehicle) through a smart contract.

The scalability of the architecture is ensured through several mechanisms. First, the integration of edge layer allows a distribution of resources at the edge of the network, thus reducing the load on the central cloud. Second, we use a dynamic resource provisioning system, where VEC nodes allocate and release processing capacities according to real-time needs. Third, the blockchain facilitates the addition of new nodes without disrupting ongoing operations, thanks to its decentralized architecture. Finally, the use of smart contracts ensures smooth automation of processes, even in cases of high demand, by dynamically adjusting service parameters and prioritizing critical transactions.

## Use Cases

### Smart City Deployment

The following use case in a smart city framework demonstrates the usefulness of our proposed architecture. We highlight in particular the effectiveness of MEC and Blockchain solutions to improve urban management and offer secure and intelligent services to users. Traffic management in a smart city presents a major challenge to optimize the flow of vehicles, increase road safety and reduce congestion. In addition, the automated management of payments related to tolls, parking or the use of urban infrastructures is crucial to guarantee public services.

The deployment of the proposed architecture makes it possible to manage data flows between vehicles and infrastructures, to automate transactions such as toll payments and to ensure the security of communications and transactions. The architecture includes a distributed Edge

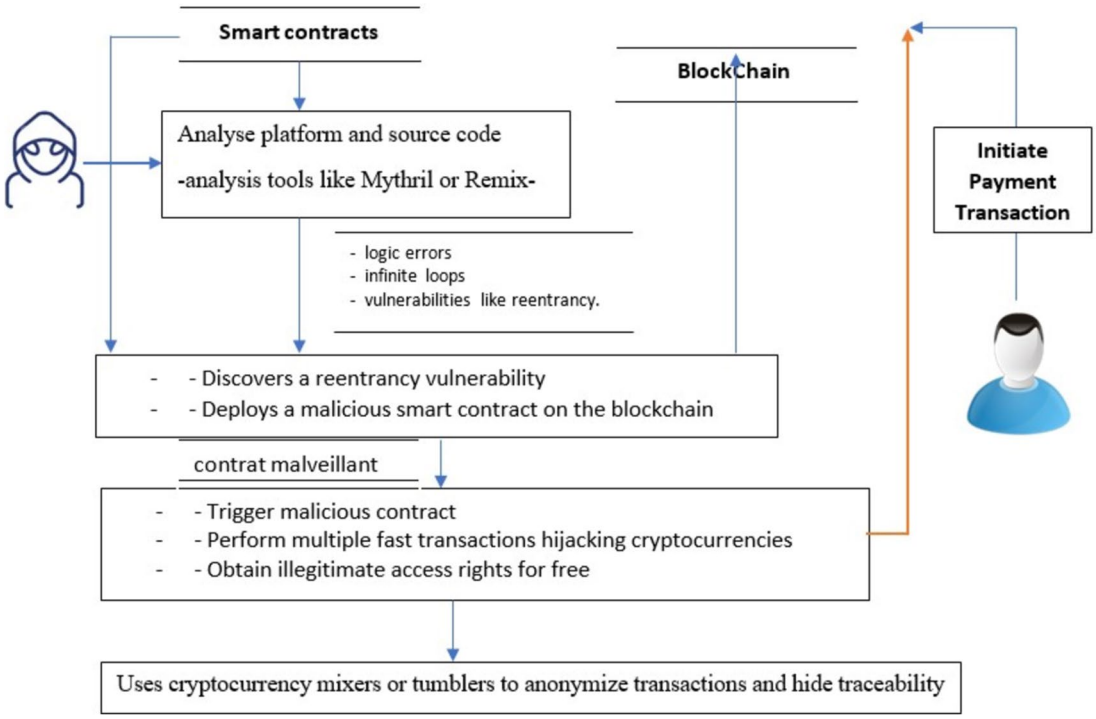



**Fig. 5** Attack scenario

computing network at the main entry and exit points of the city (e.g., intersections, toll roads, public parking lots). Each point is equipped with MEC nodes that collect information from connected vehicles and communicate with the central infrastructure via a blockchain structure. The connected vehicle is equipped with a module to communicate traffic data (e.g., position, speed, intention to park, etc.) to the MEC nodes. Each MEC node located near a road or parking lot, it locally processes the vehicle information and interacts with the blockchain to record transactions. The blockchain allows to secure and automate payments, such as toll and parking fees, via smart contracts.

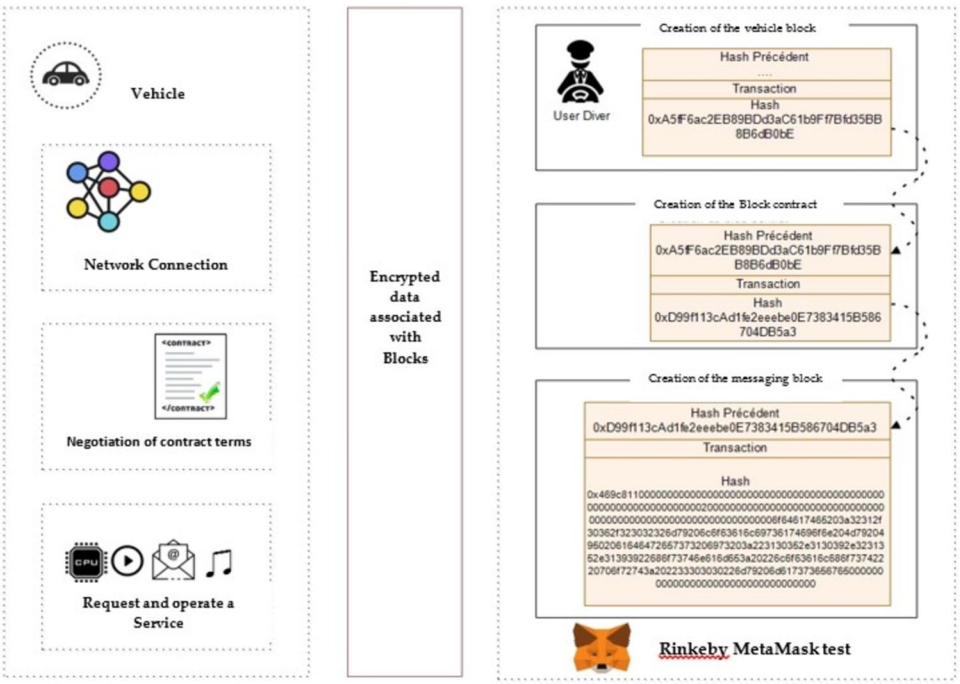



**Fig. 6** Components and Actors of VEC Systems

The objective of deploying this architecture is to improve traffic management and automate transactions by relying on blockchain and edge computing technologies. On the one hand, MEC nodes located at strategic points collect data from connected vehicles in real time, such as their position and speed. This information is then processed locally to make quick decisions, in order to avoid traffic and road incidents. On the other hand, the automation of transactions when moving a vehicle is ensured by an intelligent contact, as in the case of toll payment. In which, the details of this transaction, such as the amount paid and the time of passage, are recorded securely and transparently in the blockchain, ensuring the traceability and inviolability of the data. At the same time, vehicle drivers receive personalized services, such as notifications on the availability of parking spaces and traffic conditions, while benefiting from automated payment management.

contracts, which govern the terms agreed upon between customers and content providers.

This type of attack exploits a flaw in a smart contract responsible for managing payments and licenses on a decentralized content distribution platform. The attacker manipulates the contract to divert funds or illegally access premium content. Payments are made in cryptocurrencies, while licenses and copyrights are recorded and controlled via the blockchain, ensuring transparency and traceability. However, a vulnerability, such as a reentrancy attack, allows an attacker to siphon funds by forcing the contract to execute the same transaction multiple times before the balance is updated. Other types of attacks can target unauthorized access to licenses, creating flaws in rights and subscription management. The consequences of this type of attack are multiple:

**Algorithm 3** Smart Contract Management (V, F, S, QoS, Duration)

```
SC: SMART CONTRACT - BC: BLOCKCHAIN
if (V is Connected) then
    Set transaction terms (QoS, Price, Duration)
    Create SC ()
    Insert SC (BC)
else
    Release (S, F)
    Repeat SC (QoS, Duration)
end if
if (Duration Elapsed) then
    Release (S, F, Rating)
else
    if (QoS Decreases) then
        Refund (V)
        Penalize (F, Rating)
    end if
end if
```

## Smart Contract Attack Scenario

In the attack scenario of the Fig. 5, the objective is to exploit a vulnerability within the smart contract responsible for managing payments and licenses, with the intent of diverting funds or unlawfully accessing premium content. Specifically, the target is a decentralized content delivery platform that leverages blockchain technology to guarantee transparency, authenticity, and transaction integrity among content creators, distributors, and consumers. Payments are conducted in cryptocurrencies, while copyrights and content licenses are tracked and enforced through blockchain-based smart

- Financial loss: Users and the platform suffer significant losses due to the embezzled funds.
- Degradation of trust: The incident harms the reputation of the platform, impacting user loyalty.
- Violation of privacy: If the attacker gains access to sensitive transaction or user data, this could lead to personal data leaks.

The integration of Fog Computing and Edge Computing helps secure and optimize blockchain-based content platforms. These technologies decentralize data processing, improve performance by reducing latency, and enhance

security by processing information closer to its source. The details of fog/edge computing enhancement solutions are presented as follows:

- **Enhanced security**: Fog and Edge nodes monitor transactions in real-time and quickly detect suspicious behavior (such as a reentrancy attack) using locally deployed AI algorithms. By distributing processing across multiple nodes, the platform also becomes less vulnerable to DDoS attacks.
- **Optimized performance**: Edge nodes handle some operations locally, reducing transaction latency, such as micropayments or logging. Additionally, these nodes can cache frequently requested content, improving platform responsiveness.
- **Increased reliability**: Critical functionality, such as transaction validation, can be distributed across fog nodes to reduce reliance on a central server and ensure service continuity in the event of an outage.
- **Enhanced data protection**: Sensitive information can be processed locally by Edge nodes, limiting its exposure on the network. Data transmitted to the blockchain can also be encrypted, and encryption keys distributed across multiple nodes, making it more difficult for attackers to access it.

## Implementation Scenarios

This case study aims to show the feasibility of integrating Blockchain technology, Fog Computing, and Edge Computing in a 6G architecture. Thus, demonstrates the correct way for an efficient use of Blockchain technology in a multi-level 6G architecture based on Edge and Fog Computing. This is done through the integration of SMART Contracts to preserve users' rights (here, vehicles) and ensure a good customer/supplier relationship is managed by compliance with the terms of the contracts.

We implemented our Ethereum platform system to validate our proposal and executed various scenarios. As shown in Fig. 6, the vehicle is the primary actor on our platform; it acts as a node connected to the MetaMask blockchain test network (we utilized the Rinkeby version).

Based on the implemented platform, the user can request a service from the providers registered in the platform. The process of using a service begins with a protocol to establish a connection at the request of the vehicle; in fact, the sub-net assigns a block and a hash and then integrates it into this blockchain. After establishing the connection, several scenarios are possible:

- Once the connection is established, the vehicle is faced with a range of services (free and paid) offered by providers with an initial level of trust, the vehicle only has to choose the provider from the list of choices, and then it invokes one of the services that appears among the services offered;
- The choice of a service is necessarily followed by a negotiation of its characteristics, which must be agreed between the vehicle and the provider to build a smart contract ensuring the proper exploitation of the resources;
- Once the terms of the contract are defined and validated by both parties (requester, provider), a request to establish a contract can be transmitted by the vehicle to the Ethereum MetaMask network which assigns it a block and a hash;
- A paid service is always provided following a payment made when establishing the contract, the applicant will be notified by a message indicating the amount debited from his account;
- Once the contract is established, the exploration of the service begins, and since we are dealing with a smart contract, if at least one of the terms of the contract is not fulfilled, a forced cancellation is automatically triggered and the vehicle will be notified, and the vehicle can move to another supplier;
- In the absence of a forced cancellation of a contract, the supplier must be reimbursed and an alert indicating the amount paid to him is displayed and the requester receives a notification of the normal end of the contract;
- The requester evaluates a supplier depending on how the contract is terminated; in case of forced cancellation the level of trust will be lowered, however, it is maintained or improved during the normal termination of the contract;
- After an exploration of the platform, some disconnection is inevitable, this occurs following the request of the vehicle;
- The vehicle can interact with nodes of the same network via the free messaging service which ensures V2V communication with exchanges between vehicles concerning position coordinates, geolocation (Latitude, Longitude), IP address and others.

The proposed solution provides a sophisticated and intelligent platform prioritizing QoS for VEC systems. On the one hand, it is a flexible solution that ensures non-falsifiable transactions by utilizing smart contracts. This includes considering and managing user preferences, manipulating services, negotiating contract terms and managing user profiles in service adaptation. On the other hand, it is an intelligent solution that offers decentralized, automated and parameterized control and monitoring without needing third-party intervention, thanks to the blockchain component. The architecture of VEC systems

prioritizes QoS by dynamically allocating bandwidth and employing latency-sensitive routing protocols. Furthermore, the architecture ensures the reliability and security of V2X communications through encryption, authentication, redundancy, resilience and intrusion detection/prevention mechanisms. By optimizing resource allocation and fostering a secure communication environment, the architecture enables efficient, reliable and secure V2X communications, thereby facilitating the seamless exchange of critical information between vehicles and their surroundings.

In this study, we did not specify transmission or bandwidth protocols, opting instead for protocols tailored to the advanced capabilities of 6G networks, such as NR-U, or refined iterations of established protocols. Our approach allows us to harness the inherent efficiencies and optimizations designed for the next generation of wireless communication. Furthermore, our dynamic bandwidth allocation mechanism operates with QoS requirements, ensuring adaptable and responsive network resource utilization to meet diverse application needs.

The proposed solution enables, among other things, the following:

- Local information transfer between neighbouring MEC servers bypasses the central network and enables quick resource access.
- Local processing of collected information significantly reduces the computational burden on the central network.
- Lower download latency for users of high-speed mobile services due to reduced packet waiting times, ensuring that the link network handles less communication traffic.
- MEC platform services can provide V2X applications with essential information, allowing them to make use of radio network information (RNI) and offer services more efficiently.
- Facilitated mobile device localization and development of user-centric applications.
- Prediction of achievable QoS for each node, improving the selection process for the serving cell and MEC server by referring to the trust factor associated with each service provider based on user experience. QoS prediction also enhances the performance of specific V2X applications running on an MEC host.
- Current generations of cellular technology do not fully support the significant contribution of MEC to the development of V2X applications.
- MEC usefulness extends beyond critical V2X use cases. Storing information in local data caches enables convenient services, such as software and local traffic updates, which are only partially supported by current network architectures.
- The significant facilitation of embedded infotainment services by MEC is expected to become more advanced as the industry moves towards fully autonomous vehicles.
- Enhancement of user experience and network utilization, thus ensuring optimal QoE and seamless service coverage throughout the vehicle trajectory.

Blockchains offer numerous economic benefits for network providers and third parties, such as equipment manufacturers, software and hardware providers and other companies related to the automotive industry.

## Discussion and Challenges

In implementing the proposed VEC system, we are mindful of potential challenges and have devised mitigation strategies to address them effectively. For instance, challenges such as resource constraints and interoperability issues are tackled through collaborative planning and adherence to standardization protocols. Security concerns are mitigated through robust encryption and continuous monitoring. Furthermore, dynamic resource provisioning and capacity planning strategies are implemented to manage scalability challenges efficiently, ensuring smooth system operation even under fluctuating workloads.

Ensuring interoperability with existing communication protocols is paramount for the success of the VEC system within the broader 6G network environment. Our architecture emphasizes compliance with standardization protocols, API integration, and protocol translation mechanisms to achieve this objective. By aligning with evolving 6G standards and promoting compatibility across diverse network domains, we ensure seamless integration of the VEC system into existing communication infrastructures. This approach fosters collaboration and synergy, enabling the realization of the full potential of the 6G network environment.

Latency and bandwidth improvements are facilitated by a multi-faceted approach that integrates edge processing, resource allocation optimization, and spectrum efficiency. Leveraging fog computing and MEC strategies significantly curtails processing delays, thereby enhancing overall network responsiveness. Additionally, the implementation of blockchain technology ensures the integrity and efficiency of data transactions, bolstering the network's security and performance aspects. This comprehensive framework augments network capabilities and fosters a robust ecosystem poised for sustained advancement in next-generation communications.

Energy efficiency is a core consideration when designing our VEC system, especially considering its resource-intensive computational and storage capacities. Through edge computing optimization, blockchain pruning, and dynamic

resource allocation, we prioritize energy-efficient operation without compromising performance. Integrating low-power hardware further enhances energy efficiency, maximizing sustainability and reducing environmental impact while delivering robust performance within 6G networks.

Reducing energy consumption is a key aspect of our architecture. We use blockchain pruning techniques to reduce the size of the blockchains stored on VEC nodes. By eliminating obsolete transactions and data, we minimize the processing load and memory usage, which directly reduces the energy consumption of the equipment. In a toll management scenario, only transactions from the last 24 h are maintained on VEC nodes near the roads, as they are most likely to be accessed for verification or dispute. Older transactions are transferred to the cloud, where they can be archived for legal reference without straining local resources. This saves VEC nodes from continuously processing and storing a large chain, thereby reducing the energy consumption of onboard computing devices.

In parallel, our system relies on dynamic resource allocation. MEC nodes only activate their processing capacities when there is real demand, which allows servers to be put to sleep when activity is low. In a VEC system for traffic management, VEC nodes activate their processing capabilities only during peak periods, such as in the morning and at the end of the day. During these periods, the nodes process V2X information in real time to adjust traffic lights and optimize traffic. However, during nighttime or outside peak hours, VEC nodes automatically disable some non-critical features, such as infotainment services, and switch to low-power mode. This adaptive management ensures energy optimization without compromising Qos.

## Conclusion

In this paper, we proposed a novel architecture that integrates Blockchain and Edge Computing paradigms to enhance security and resource management in 6G networks. Focusing on VEC, our goal was to preserve QoS and QoE for end-users by leveraging Smart Contracts to ensure the non-repudiation of transactions. The inherent security and privacy benefits of Blockchain technology are directly inherited within the introduced framework. Furthermore, Bringing cloud resources closer to the network edge minimizes latency and meets the exponentially growing demand for resources driven by real-time system status updates.

In our architecture, objects compete to maximize resource utilization efficiency and meet QoS requirements to achieve their objectives. Smart Contracts provide absolute trust in the execution of agreements in our platform compared to others. The agreements' transparency, autonomy, and security eliminate any attempts at bias, manipulation, or error.

Among future perspectives, several avenues deserve exploration to complement and extend this work. First, integrating decision support systems for optimal provider selection based on evaluations can enable fully autonomous VEC communications, eliminating human dependency. Second, dynamic contract renegotiation during the time of execution can be explored. Third, a recommendation system could be implemented to provide customized and personalized services tailored to requester profiles. Finally, the inherent traceability of Blockchain can be leveraged to extract vehicle routes using the platform's existing geolocation functionality for security purposes.

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